THE QUIVER ASSOCIATED TO THE CATEGORY OF G_n-EQUIVARIANT PERVERSE SHEAVES ON $\mathrm{M}_n(\mathbb{C})$

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1. Introduction

Many categories of perverse sheaves can be described as categories of representations of quivers with relations [GMV96, Vil94], although explicit descriptions are often difficult to obtain.

This article treats one of them. Let $\mathrm{X}_n := \mathrm{M}_n(\mathbb{C})$ be the variety of $n \times n$ complex matrices and let $\mathrm{G}_n := \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C})$ act on X_n by $(g, h) \cdot x = gxh^{-1}$. The orbits of this action are the sets X_n^k of matrices of corank k, for $0 \leq k \leq n$, which gives a stratification of the variety X_n. Braden and Grinberg [BG99] studied perverse sheaves with respect to this stratification with coefficients in $\mathbb{C}$. Later, and in a different but parallel setting, Lőrincz and Walther [LW19] showed that the category of G_n-equivariant coherent $\mathcal{D}$-modules on X_n is equivalent to the category of representations of the quiver $\widehat{AA_n}$:

$$0 \underset{v_0}{\overset{u_0}{\rightleftarrows}} 1 \underset{v_1}{\overset{u_1}{\rightleftarrows}} 2 \underset{v_2}{\overset{u_2}{\rightleftarrows}} \cdots \underset{v_{n-1}}{\overset{u_{n-1}}{\rightleftarrows}} n$$

with the relations

$$u_i v_i = 0 = v_i u_i, \qquad 0 \leq i \leq n-1.$$

Let $\Bbbk$ be a field. Denote the category of G_n-equivariant perverse sheaves on X_n with coefficients in $\Bbbk$ by $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$. It is a natural question to ask whether the quiver description exists for $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$. Cipriani and Woolf [CW22] show the quiver description exists as long as there are finitely many strata and the category of local systems on each of them has enough projectives, which is true in our case.

We show that there is a unique simple object associated to each stratum, which we denote by IC_k for $0 \leq k \leq n$. Since the number of vertices is the number of simple objects, we get $n+1$ vertices of the quiver. The dimension of Ext^1 groups between the simple objects gives us all the arrows. The relations between the arrows are not easy to obtain and usually need information about multiplicities of simples in projective covers. In our case, we will show how to get all the relations inductively. This gives the following main result:

Theorem 1.1. *Let $\Bbbk$ be a field. For every $n \geq 1$ there is an equivalence of categories*

$$\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk) \cong \mathrm{rep}(\widehat{AA_n}, \Bbbk).$$

The Theorem 1.1 lets one transport structural facts about $\mathrm{rep}(\widehat{AA_n})$, where everything is computable by hand, back to the geometric side.

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1.1. **Applications.** The Fourier–Sato transform $\mathbb{T}$ is a t-exact autoequivalence of the category $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$ that carries the constant sheaf IC_0 to the skyscraper IC_n. Its effect on the other simple objects is harder to compute in general. We will show how the uniseriality of composition factors in projective covers of IC_0 gives

$$\mathbb{T}(\mathrm{IC}_k) \cong \mathrm{IC}_{n-k}, \qquad 0 \leq k \leq n.$$

Secondly, it's easier to construct the projective cover of simple objects from the quiver side. Thus, we can construct the projective cover of IC_k where each simple occurs exactly once. For $0 \leq k \leq n$, we will show that the perverse sheaf

$${}^{p}\mathrm{H}^0(j^k_{0!}j_0^{k*}(\mathbb{T}({}^{p}\mathrm{H}^0(j_{n-k!}\Bbbk[\dim \mathrm{X}_n^{n-k}]))))$$

is the projective cover of IC_k in $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$, where $j_k : \mathrm{X}_n^k \hookrightarrow \mathrm{X}_n$ and $j_0^k : \cup_{i=0}^k \mathrm{X}_n^i \hookrightarrow \mathrm{X}_n$ are the inclusion maps.

2. Preliminaries

Let $\mathrm{X}_n := \mathrm{M}_n(\mathbb{C})$ and $\mathrm{G}_n := \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C})$ be the group acting on X_n by the following action

$$(g, h) \cdot x = gxh^{-1}$$

for all $(g, h) \in \mathrm{G}_n$ and $x \in \mathrm{X}_n$. The orbits of this action are completely classified by the corank of the matrix. This gives a stratification of X_n in terms of corank of matrices, i.e.,

$$\mathrm{X}_n = \bigsqcup_{0 \leq k \leq n} \mathrm{X}_n^k$$

where $\mathrm{X}_n^k = \{x \in \mathrm{X}_n \mid \mathrm{corank}(x) = k\}$.

Proposition 2.1. *For $0 \leq k \leq n$, we have:*

$$\overline{\mathrm{X}_n^k} = \bigsqcup_{l \geq k} \mathrm{X}_n^l$$

Proof. For $0 \leq k \leq n$, let

$$x_k = \begin{pmatrix} I_{n-k} & 0 \\ 0 & 0 \end{pmatrix}.$$

Let $l > k$. Any matrix $A \in \mathrm{X}_n^l$ is of the form $g \cdot x_l$ for some $g \in \mathrm{G}_n$. Consider the well-defined map

$$\begin{aligned} \varphi_g : \mathbb{C} &\to \mathrm{X}_n \\ t &\mapsto g \cdot \begin{pmatrix} I_{n-l} & 0 & 0 \\ 0 & t\, I_{l-k} & 0 \\ 0 & 0 & 0_k \end{pmatrix}. \end{aligned}$$

For $t \neq 0$, we have $\mathrm{corank}(\varphi_g(t)) = k$ i.e. $\varphi_g(t) \in \mathrm{X}_n^k$ and $\varphi_g(0) = A$. Since φ_g is continuous, $\varphi_g^{-1}\big(\overline{\mathrm{X}_n^k}\big)$ is closed in $\mathbb{C}$ and contains the dense subset $\mathbb{C}^\times$, hence

$$\varphi_g^{-1}\big(\overline{\mathrm{X}_n^k}\big) = \mathbb{C}.$$

Thus, $A \in \overline{\mathrm{X}_n^k}$. Since A was arbitrarily chosen, we get $\mathrm{X}_n^l \subset \overline{\mathrm{X}_n^k}$ for all $l > k$. Clearly, $\mathrm{X}_n^k \subset \overline{\mathrm{X}_n^k}$ and we get

$$\bigsqcup_{l \geq k} \mathrm{X}_n^l \subset \overline{\mathrm{X}_n^k}.$$

Since $\bigsqcup_{l\geq k} X_n^l$ is a Zariski closed variety defined by the vanishing of $(n-k+1)\times(n-k+1)$ minors in X_n, we get our result. □

Lemma 2.2. *The stabilizer of any matrix $x \in X_n$ is connected.*

Proof. The stabilizer of $x \in X_n$ is the subgroup:

$$\mathrm{Stab}(x) = \{(g,h) \in G_n \mid gx = xh\}$$

Suppose the matrix x has corank r. Every matrix of corank r lies in the G_n-orbit of

$$x_r = \begin{pmatrix} I_{n-r} & 0 \\ 0 & 0 \end{pmatrix}$$

where I_{n-r} is the identity matrix of size $n-r$. Since stabilizers of points in one orbit are conjugate, it suffices to show $\mathrm{Stab}(x_r)$ is connected.

Let $g, h \in \mathrm{GL}_n(\mathbb{C})$ be partitioned into the same block structure

$$g = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad h = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$$

where A and P are $(n-r)\times(n-r)$ blocks. The stabilizer condition $gx_r = x_r h$ implies $A = P, C = 0$ and $Q = 0$.

Since g and h are invertible, their diagonal blocks A, D, and S must also be invertible. Thus, the stabilizer of x_r takes the form:

$$\mathrm{Stab}(x_r) = \left\{ \left(\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \begin{pmatrix} A & 0 \\ R & S \end{pmatrix} \right) \;\middle|\; \begin{array}{l} A \in \mathrm{GL}_{n-r}(\mathbb{C}),\ D, S \in \mathrm{GL}_r(\mathbb{C}), \\ B, R^T \in \mathrm{M}_{n-r,r}(\mathbb{C}) \end{array} \right\}. \tag{1}$$

The Levi factor consisting of the block diagonal pairs determined by (A, D, S) forms a subgroup isomorphic to $\mathrm{GL}_{n-r}(\mathbb{C}) \times \mathrm{GL}_r(\mathbb{C}) \times \mathrm{GL}_r(\mathbb{C})$. The unipotent radical consists of the pairs parameterized by B and R; it is a normal subgroup isomorphic to the additive group of complex matrices $\mathrm{M}_{n-r,r}(\mathbb{C}) \oplus \mathrm{M}_{r,n-r}(\mathbb{C})$. Since the stabilizer is a semidirect product of two connected groups, the entire stabilizer group $\mathrm{Stab}(x_r)$ is connected. □

The above explicit description of stabilizers can help us to find the dimension of our strata. By the orbit–stabilizer theorem

$$\dim X_n^k = \dim G_n - \dim \mathrm{Stab}(x_k) = 2n^2 - \dim \mathrm{Stab}(x_k).$$

By Lemma 2.2 (with $r = k$), the stabilizer group $\mathrm{Stab}(x_k)$ is the semidirect product of the Levi factor $\mathrm{GL}_{n-k}(\mathbb{C}) \times \mathrm{GL}_k(\mathbb{C}) \times \mathrm{GL}_k(\mathbb{C})$, parametrized by (A, D, S), and the unipotent radical $\mathrm{M}_{n-k,k}(\mathbb{C}) \oplus \mathrm{M}_{k,n-k}(\mathbb{C})$, parametrized by (B, R). Hence

$$\dim \mathrm{Stab}(x_k) = \underbrace{(n-k)^2 + 2k^2}_{\text{Levi}} + \underbrace{2k(n-k)}_{\text{unipotent}} = (n-k)(n+k) + 2k^2 = n^2 + k^2.$$

Therefore,

$$\dim X_n^k = 2n^2 - (n^2 + k^2) = n^2 - k^2.$$

Moreover, we can also calculate

$$\dim \overline{X_n^k} = \max_{l\geq k}\{\dim X_n^l\} = n^2 - k^2 = \dim X_n^k\,. \tag{2}$$

Let $\Bbbk$ be a field and $\mathrm{D}^b_{G_n}(X_n, \Bbbk)$ be the bounded derived category of G_n-equivariant complexes on X_n with coefficients in $\Bbbk$. Let $\mathrm{Perv}_{G_n}(X_n, \Bbbk) \subset \mathrm{D}^b_{G_n}(X_n, \Bbbk)$ be the full subcategory of G_n-equivariant perverse sheaves on X_n with

coefficients in $\Bbbk$. Moreover, we will denote the constant sheaf on any space X by $\underline{\Bbbk}_{\mathrm{X}}$.

Local systems are the building blocks of perverse sheaves. The following lemma describes the category of local systems on each stratum X_n^k.

Lemma 2.3. *For $0 \leq k \leq n$, the category of local systems on X_n^k i.e., $\mathrm{Loc}_{\mathrm{G}_n}(\mathrm{X}_n^k, \Bbbk)$ is equivalent to the category of $\Bbbk$-vector spaces.*

Proof. Each stratum is a homogeneous G_n-space with connected stabilizers using the Lemma 2.2, so by [Ach21, Prop. 6.2.13], there are no nontrivial G_n-equivariant local systems on X_n^k. □

Since vector spaces are projective, the equivariant category of local systems on each stratum has enough projectives. This information is important for our next lemma.

Lemma 2.4. *The category $\mathrm{Perv}_{\mathrm{G}_n}(X_n, \Bbbk)$ has enough projectives.*

Proof. Let

$$\mathrm{X}_n^k \hookrightarrow \overline{\mathrm{X}_n^k},$$
$$\overline{\mathrm{X}_n^{k-1}} \hookrightarrow \overline{\mathrm{X}_n^k}$$

be the open and the complementary closed embeddings. These induce a recollement of equivariant derived categories

$$\mathrm{D}^b_{\mathrm{G}_n}(X_n^k, \Bbbk) \longrightarrow \mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^k}, \Bbbk) \longrightarrow \mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^{k-1}}, \Bbbk),$$

as in [BBDG18, §1.4.3].

For $0 \leq k \leq n$, the heart of the category $\mathrm{D}^b_{\mathrm{G}_n}(\mathrm{X}_n^k, \Bbbk)$ is

$$\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n^k, \Bbbk) \cong \mathrm{Loc}_{\mathrm{G}_n}(\mathrm{X}_n^k, \Bbbk)[\dim \mathrm{X}_n^k],$$

which is a length category with finitely many simple objects. Using the Lemma 2.3, it has enough projectives. Moreover, the heart of the category $\mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^k}, \Bbbk)$ is $\mathrm{Perv}_{\mathrm{G}_n}(\overline{\mathrm{X}_n^k}, \Bbbk)$, which is again a length category with finitely many simple objects for all $0 \leq k \leq n$.

We will use induction on k to show our result. For $k = 0$, the heart of $\mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^0}, \Bbbk)$ is

$$\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n^0, \Bbbk) \cong \mathrm{Loc}_{\mathrm{G}_n}(\mathrm{X}_n^0, \Bbbk)[\dim \mathrm{X}_n^0]$$

and has enough projectives.

Using induction on k and [CW22, Proposition 4.8], the heart of the glued t-structure on $\mathrm{D}^b_{G_n}(\overline{\mathrm{X}_n^k}, \Bbbk)$, namely $\mathrm{Perv}_{\mathrm{G}_n}(\overline{\mathrm{X}_n^k}, \Bbbk)$, has enough projectives. Taking $k = n$ gives the lemma. □

When there are enough projective perverse sheaves and finitely many simple objects, the category of perverse sheaves can be described as modules over a finite-dimensional algebra, and therefore also as a category of representations of a quiver with relations. The rest of the story is the quest to find this quiver.

3. Building the quiver

We establish some notation and review some basic results regarding the representation theory of quivers, following [ASS06]. A *quiver* Q is an oriented graph, i.e. a pair $Q = (Q_0, Q_1)$ formed by a finite set of vertices Q_0 and a finite set of arrows Q_1. An arrow $a \in Q_1$ has a head (or target) ha and a tail (or source) ta which are elements of Q_0:

$$ta \xrightarrow{a} ha$$

The $\Bbbk$-vector space with basis given by the (directed) paths in Q has a natural multiplication induced by concatenation of paths. The corresponding $\Bbbk$-algebra is called the *path algebra* of the quiver Q and is denoted $\Bbbk Q$.

A *relation* in Q is a $\Bbbk$-linear combination of paths of length at least two having the same source and target. We define a *quiver with relations* $\widehat{Q} := (Q, I)$ to be a quiver Q together with a finite set of relations I. The *quiver algebra* of $\widehat{Q}$ is the quotient $\Bbbk Q/\langle I\rangle$ of the path algebra by the ideal generated by the relations. We always assume that the ideal of relations $\langle I\rangle$ contains any path whose length is large enough, so that the corresponding quiver algebra is finite-dimensional (see [ASS06, Section II.2]). We will often use the word quiver to refer to a quiver with relations.

A (finite-dimensional) representation V of a quiver $\widehat{Q}$ is a family of (finite-dimensional) vector spaces $\{V_x \mid x \in Q_0\}$ together with linear maps $\{V(a)\colon V_{ta} \to V_{ha} \mid a \in Q_1\}$ satisfying the relations induced by the elements of I. A morphism $\phi\colon V \to V'$ of two representations V, V' of $\widehat{Q}$ is a collection of linear maps $\phi = \{\phi(x)\colon V_x \to V'_x \mid x \in Q_0\}$, such that for each $a \in Q_1$ we have $\phi(ha) \circ V(a) = V'(a) \circ \phi(ta)$. We note that the data of a representation of $\widehat{Q}$ is equivalent to that of a module over the quiver algebra, and in fact, the category $\operatorname{rep}(\widehat{Q})$ of finite-dimensional representations of $\widehat{Q}$ is equivalent to that of the finitely generated $\Bbbk Q/\langle I\rangle$-modules [ASS06, Section III.1, Thm. 1.6]. This category is Abelian, Artinian, Noetherian, has enough projectives and injectives, and contains only finitely many simple objects, seen as follows.

The (isomorphism classes of) simple objects in $\operatorname{rep}(\widehat{Q})$ are in bijection with the vertices of Q. For each $x \in Q_0$, the corresponding simple S^x is the representation with $(S^x)_x = \Bbbk$, $(S^x)_y = 0$ for all $y \in Q_0 \setminus \{x\}$. A (non-zero) representation of $\widehat{Q}$ is called *indecomposable* if it is not isomorphic to a direct sum of two non-zero representations. For each $x \in Q_0$, we let P^x (resp. I^x) denote the *projective cover* (resp. *injective envelope*) of S^x, as constructed in [ASS06, Section III.2]. In particular, for $y \in Q_0$, the dimension of $(P^x)_y$ (resp. $(I^x)_y$) is given by the number of paths from x to y (resp. from y to x), considered up to the relations in I.

Given two quivers with relations, $\widehat{Q} = (Q, I)$ and $\widehat{Q}' = (Q', I')$, we say that $\widehat{Q}'$ is a *subquiver* of $\widehat{Q}$, if Q' is an (oriented) subgraph of Q, and the relations generated by I' contain the relations from I that are induced from Q to Q'. Clearly, this gives a surjective map $\Bbbk Q/\langle I\rangle \to \Bbbk Q'/\langle I'\rangle$ and $\operatorname{rep}(\widehat{Q}')$ is naturally a full subcategory of $\operatorname{rep}(\widehat{Q})$.

We use this to recover the quiver from the category of perverse sheaves. The simple objects of the category will be in one-to-one correspondence with the vertices of the quiver. Moreover, the number of arrows from vertex x to vertex y will be calculated by finding the dimension of the first extension group, i.e., $\dim_{\Bbbk} \operatorname{Ext}^1(S^x, S^y)$

(from [ASS06, Lemma 2.12]), where S^x and S^y are the simple perverse sheaves corresponding to the vertices x and y respectively.

3.1. **The vertices.** The simple objects in $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk) \subset \mathrm{D}^b_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$ are the complexes $\mathrm{IC}(\mathrm{X}^k_n, L)$ where L runs over irreducible G_n-equivariant local systems on X^k_n. Using the Lemma 2.3, the only such irreducible local system is the trivial local system.

Therefore,

$$\mathrm{IC}_k := \mathrm{IC}(\mathrm{X}^k_n, \underline{\Bbbk}_{\mathrm{X}^k_n}[\dim \mathrm{X}^k_n])$$

for $0 \leq k \leq n$, are all the simple objects up to isomorphism in $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$. This also tells us that there are $n+1$ vertices in the quiver associated to $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$.

$$0 \qquad 1 \qquad 2 \qquad \cdots \qquad n$$

The vertex k in the above quiver corresponds to the simple perverse sheaf IC_k.

We will give an alternative description of the IC_k which will help us compute their stalks. Let

$$\widetilde{\mathrm{X}^k_n} = \{(x, W) \in \overline{\mathrm{X}^k_n} \times \mathrm{Gr}(k, n) \mid W \subset \ker(x)\}$$

where $\mathrm{Gr}(k, n)$ is the Grassmannian variety of k-dimensional spaces in $\mathbb{C}^n$. Let G_n act on $\widetilde{\mathrm{X}^k_n}$ by

$$(g_1, g_2) \cdot (x, W) = (g_1 x g_2^{-1}, g_2 \cdot W)$$

for all $(g_1, g_2) \in \mathrm{G}_n$. Under the usual $\mathrm{GL}_n(\mathbb{C})$-action on $\mathrm{Gr}(k, n)$, the second projection $p_k : \widetilde{\mathrm{X}^k_n} \to \mathrm{Gr}(k, n)$ is $\mathrm{GL}_n(\mathbb{C})$-equivariant, and the fibers are linear and of the same dimension. This makes p_k a vector bundle over a smooth variety. Thus, $\widetilde{\mathrm{X}^k_n}$ is a smooth variety for all k.

In fact, each G_n-orbit of $\widetilde{\mathrm{X}^k_n}$ is open in its closure, hence a smooth locally closed subvariety. The orbits of $\widetilde{\mathrm{X}^k_n}$ are of the form $\pi_k^{-1}(\mathrm{X}^l_n)$ where $l \geq k$ and π_k is the first projection map. Thus, we get the following stratification:

$$\widetilde{\mathrm{X}^k_n} = \bigsqcup_{l \geq k} \pi_k^{-1}(\mathrm{X}^l_n).$$

Lemma 3.1. *The first projection map $\pi_k : \widetilde{\mathrm{X}^k_n} \to \overline{\mathrm{X}^k_n}$ is proper and small with respect to X^k_n.*

Proof. Consider the first projection map $\Pi_k : \overline{\mathrm{X}^k_n} \times \mathrm{Gr}(k, n) \to \overline{\mathrm{X}^k_n}$. Since $\mathrm{Gr}(k, n)$ is a projective variety, Π_k is proper. Moreover, $\widetilde{\mathrm{X}^k_n}$ is a closed subvariety of $\overline{\mathrm{X}^k_n} \times \mathrm{Gr}(k, n)$ which implies the restriction $\pi_k = \Pi_k|_{\widetilde{\mathrm{X}^k_n}}$ is proper.

The biggest stratum of $\widetilde{\mathrm{X}^k_n}$ is

$$\pi_k^{-1}(\mathrm{X}^k_n) = \{(x, W) \in \mathrm{X}^k_n \times \mathrm{Gr}(k, n) \mid W \subset \ker(x)\} = \{(x, \ker(x)) \mid x \in \mathrm{X}^k_n\} \cong \mathrm{X}^k_n\,.$$

Using 2, we get

$$\dim\ \widetilde{\mathrm{X}^k_n} = n^2 - k^2.$$

For $l \geq k$ and $x_n^l \in \mathrm{X}_n^l$, consider the fiber over x_n^l, i.e., $\pi_k^{-1}(x_n^l)$. Since the dimension of $\ker(x_n^l) = l$, we get

$$\begin{aligned}\pi_k^{-1}(x_n^l) &= \{(x_n^l, W) \mid W \subset \ker(x_n^l) \text{ and } W \in \mathrm{Gr}(k,n)\} \\ &= \{(x_n^l, W) \mid W \in \mathrm{Gr}(k,l)\} \cong \mathrm{Gr}(k,l),\end{aligned}$$

which gives

$$\dim\ \pi_k^{-1}(x_n^l) = \dim\ \mathrm{Gr}(k,l) = k(l-k).$$

Note that X_n^k is an open dense subset in $\overline{\mathrm{X}_n^k}$ and for $x_n^l \in \mathrm{X}_n^l$ where $l > k$,

$$\dim\ \pi_k^{-1}(x_n^l) = k(l-k) < \frac{1}{2}(l^2 - k^2) = \frac{1}{2}(\dim\ \widetilde{\mathrm{X}_n^k} - \dim\ \mathrm{X}_n^l).$$

Thus, the map $\pi_k : \widetilde{\mathrm{X}_n^k} \to \overline{\mathrm{X}_n^k}$ is small with respect to X_n^k. □

Corollary 3.2. *For $0 \leq k \leq n$,*

$$\mathrm{IC}_k \cong \pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim\ \mathrm{X}_n^k]. \tag{3}$$

Proof. Since the maps π_k are small with respect to X_n^k and

$$\pi_k^{-1}(\mathrm{X}_n^k) = \mathrm{X}_n^k \times \mathrm{Gr}(k,k) \cong \mathrm{X}_n^k,$$

the result follows from [Ach21, Prop. 3.8.10]. □

It becomes easier to calculate stalks with the above description of IC objects. The following lemma gives a local description of our simple objects. Let ${}^p\mathrm{H}^i$ and ${}^p\tau^{\leq i}$ be the functors for the perverse cohomology and truncation respectively.

Lemma 3.3. *For $n \geq l \geq k \geq 0$ and $t > -(l-k)^2$,*

$${}^p\mathrm{H}^t(\mathrm{IC}_k\,|_{\mathrm{X}_n^l}, \Bbbk) = 0$$
$${}^p\mathrm{H}^t(\mathrm{IC}_k\,|_{\overline{\mathrm{X}_n^l}}, \Bbbk) = 0.$$

Proof. Using the isomorphism 3, for $x_n^l \in \mathrm{X}_n^l$,

$$\mathrm{H}^t((\pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim \mathrm{X}_n^k])_{x_n^l}, \Bbbk) = \mathrm{H}^{t+\dim\ \mathrm{X}_n^k}_{singular}(\pi_k^{-1}(\mathrm{X}_n^l), \Bbbk) = \mathrm{H}^{t+\dim\ \mathrm{X}_n^k}_{singular}(\mathrm{Gr}(k,l), \Bbbk)$$

for all $k \leq l \leq n$. The top cohomology of $\mathrm{Gr}(k,l)$ lies in degree $2k(l-k)$ which implies

$$\mathrm{H}^t(\pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim\ \mathrm{X}_n^k]_{x_n^l}, \Bbbk) = 0$$

for $t > 2k(l-k) - \dim\ \mathrm{X}_n^k = -\dim\ \mathrm{X}_n^l - (l-k)^2$. This implies

$${}^p\mathrm{H}^t(\pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim\ \mathrm{X}_n^k]_{x_n^l}, \Bbbk) = \mathrm{H}^{t-\dim \mathrm{X}_n^l}(\pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim\ \mathrm{X}_n^k]_{x_n^l}, \Bbbk) = 0$$

for $t > -(l-k)^2$.

Since $\overline{\mathrm{X}_n^l} = \sqcup_{i \geq l} \mathrm{X}_n^i$, and for $i \geq l$,

$${}^p\mathrm{H}^t(\pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim\ \mathrm{X}_n^k]_{\mathrm{X}_n^i}, \Bbbk) = 0$$

for all $t > -(l-k)^2 \geq -(i-k)^2$, we get ${}^p\mathrm{H}^t(\pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim\ \mathrm{X}_n^k]_{\overline{\mathrm{X}_n^l}}, \Bbbk) = 0$ for $t > -(l-k)^2$ from [Ach21, Lemma 3.1.6]. □

In particular, for the inclusion map, $i_{l,k} : \overline{\mathrm{X}_n^l} \hookrightarrow \overline{\mathrm{X}_n^k}$, we have

$$i_{l,k}^* \,\mathrm{IC}_k \in {}^p\mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^l}, \Bbbk)^{\leq -(l-k)^2}. \tag{4}$$

Since each IC_k is Verdier self-dual, we get

$$i_{l,k}^! \,\mathrm{IC}_k \cong \mathbb{D}(i_{l,k}^* \pi_{k*} \underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim\ \mathrm{X}_n^k]) \in {}^p\mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^l}, \Bbbk)^{\geq (l-k)^2}. \tag{5}$$

3.2. **The arrows.** In this section, we use the results above to compute $\mathrm{Ext}^1(\mathrm{IC}_k, \mathrm{IC}_l)$ for $0 \leq k, l \leq n$.

Lemma 3.4. *The first G_n-equivariant cohomology of X_n vanishes, i.e.,*

$$\mathrm{H}^1_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk) = \mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}_n}, \underline{\Bbbk}_{\mathrm{X}_n}[1]) = 0.$$

Proof. Observe that X_n retracts G_n-equivariantly onto the zero matrix via the homotopy $(t, x) \mapsto tx$. Thus their equivariant cohomologies are isomorphic i.e.,

$$\mathrm{H}^1_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk) \cong \mathrm{H}^1_{\mathrm{G}_n}(\{0_n\}, \Bbbk) \cong \mathrm{H}^1(\mathrm{BG}_n, \Bbbk)$$

where BG_n is the classifying space of G_n. Since $\mathrm{G}_n = \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C})$, we get $\mathrm{BG}_n \simeq \mathrm{BGL}_n(\mathbb{C}) \times \mathrm{BGL}_n(\mathbb{C})$. From [MS74, Theorem 14.5], the cohomology ring of $\mathrm{BGL}_n(\mathbb{C})$ is a polynomial ring generated by the universal Chern classes

$$\mathrm{H}^*(\mathrm{BGL}_n(\mathbb{C}), \Bbbk) \cong \Bbbk[c_1, c_2, \ldots, c_n],$$

where the topological degree of each Chern class is even (i.e., $\deg(c_i) = 2i$).

By the Künneth formula, $\mathrm{H}^*(\mathrm{BG}_n, \Bbbk)$ is the tensor product of two such polynomial rings. Since all generators lie in even degrees, the cohomology groups in all odd degrees must vanish. Therefore, $\mathrm{H}^1_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk) = 0$. □

In a similar fashion, the same holds for each orbit X_n^k.

Lemma 3.5. *The first G_n-equivariant cohomology of X_n^k vanishes, i.e.,*

$$\mathrm{H}^1_{\mathrm{G}_n}(\mathrm{X}_n^k, \Bbbk) = \mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}_n^k}, \underline{\Bbbk}_{\mathrm{X}_n^k}[1]) = 0.$$

Proof. Using the orbit-stabilizer theorem, we have

$$\mathrm{X}_n^k \cong \mathrm{G}_n \,/\, \mathrm{Stab}(x_k).$$

Moreover,

$$\mathrm{H}^1_{\mathrm{G}_n}(\mathrm{X}_n^k, \Bbbk) \cong \mathrm{H}^1_{\mathrm{G}_n}(\mathrm{G}_n \,/\, \mathrm{Stab}(x_k), \Bbbk) \cong \mathrm{H}^1(\mathrm{BStab}(x_k), \Bbbk)$$

where $\mathrm{BStab}(x_k)$ is the classifying space of $\mathrm{Stab}(x_k)$.

From 1, the unipotent radical of $\mathrm{Stab}(x_k)$ i.e., $\mathrm{M}_{n-k,k}(\mathbb{C}) \oplus \mathrm{M}_{k,n-k}(\mathbb{C})$ is contractible. So, $\mathrm{Stab}(x_k)$ retracts to its Levi subgroup i.e., $\mathrm{GL}_{n-k} \times \mathrm{GL}_k \times \mathrm{GL}_k$. Thus,

$$\mathrm{H}^*(\mathrm{BStab}(x_k), \Bbbk) \cong \mathrm{H}^*(\mathrm{BGL}_{n-k}(\mathbb{C}), \Bbbk) \otimes \mathrm{H}^*(\mathrm{BGL}_k(\mathbb{C}), \Bbbk) \otimes \mathrm{H}^*(\mathrm{BGL}_k(\mathbb{C}), \Bbbk)$$

by the Künneth formula. Similar to Lemma 3.4, the cohomology groups survive only in even degrees. Therefore, $\mathrm{H}^1_{\mathrm{G}_n}(\mathrm{X}_n^k, \Bbbk) = 0$. □

The following lemma tells us that there are no loops in our associated quiver.

Lemma 3.6. *For $0 \leq k \leq n$,*

$$\mathrm{Ext}^1(\mathrm{IC}_k, \mathrm{IC}_k) = 0$$

Proof. Consider the distinguished triangle,

$$j_{k!}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k] \to \mathrm{IC}_k \to i_{k!}i_k^* \mathrm{IC}_k \to,$$

where $j_k : \mathrm{X}_n^k \hookrightarrow \overline{\mathrm{X}_n^k}$ is the open embedding and i_k the complementary closed embedding. Applying the functor $\mathrm{Hom}(-, \mathrm{IC}_k[1])$ and the adjoint relation, we get the following long exact sequence

$$\mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k], \underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k + 1]) \leftarrow \mathrm{Ext}^1(\mathrm{IC}_k, \mathrm{IC}_k) \leftarrow \mathrm{Hom}(i_k^* \mathrm{IC}_k, i_k^! \mathrm{IC}_k[1]) \leftarrow .$$

Using the Lemma 3.5, we have

$$\mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k], \underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k + 1]) = 0.$$

From [Ach21, Lemma 3.3.4], we can see that $i_k^* \mathrm{IC}_k \in {}^p\mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^{k+1}}, \Bbbk)^{\leq -1}$ and $i_k^! \mathrm{IC}_k[1] \in {}^p\mathrm{D}^b_{\mathrm{G}_n}(\overline{\mathrm{X}_n^{k+1}}, \Bbbk)^{\geq 0}$ which implies

$$\mathrm{Hom}(i_k^* \mathrm{IC}_k, i_k^! \mathrm{IC}_k[1]) = 0.$$

Thus,

$$\mathrm{Ext}^1(\mathrm{IC}_k, \mathrm{IC}_k) = 0$$

for all $0 \leq k \leq n$. □

The following lemma tells us that there are no arrows between non-consecutive vertices.

Lemma 3.7. *For $0 \leq k, l \leq n$, and $l - k \geq 2$,*

$$\mathrm{Ext}^1(\mathrm{IC}_k, \mathrm{IC}_l) = 0$$
$$\mathrm{Ext}^1(\mathrm{IC}_l, \mathrm{IC}_k) = 0$$

Proof. Let $i_{l,k} : \overline{\mathrm{X}_n^l} \hookrightarrow \overline{\mathrm{X}_n^k}$. Since IC_l is supported on $\overline{\mathrm{X}_n^l} \subset \overline{\mathrm{X}_n^k}$,

$$\begin{aligned} \mathrm{Ext}^1(\mathrm{IC}_k, \mathrm{IC}_l) &= \mathrm{Hom}(\mathrm{IC}_k, i_{l,k*} \mathrm{IC}_l[1]) \\ &= \mathrm{Hom}(i_{l,k}^* \mathrm{IC}_k, \mathrm{IC}_l[1]) \\ &= 0 \end{aligned} \tag{4}$$

and

$$\begin{aligned} \mathrm{Ext}^1(\mathrm{IC}_l, \mathrm{IC}_k) &= \mathrm{Hom}(i_{l,k*} \mathrm{IC}_l, \mathrm{IC}_k[1]) \\ &= \mathrm{Hom}(i_{l,k!} \mathrm{IC}_l, \mathrm{IC}_k[1]) \\ &= \mathrm{Hom}(\mathrm{IC}_l, i_{l,k}^! \mathrm{IC}_k[1]) \\ &= 0. \end{aligned} \tag{5}$$

Hence, for $0 \leq k, l \leq n$, and $l - k \geq 2$,

$$\mathrm{Ext}^1(\mathrm{IC}_k, \mathrm{IC}_l) = 0$$
$$\mathrm{Ext}^1(\mathrm{IC}_l, \mathrm{IC}_k) = 0.$$

□

Let

$$j_k : \mathrm{X}_n^k \hookrightarrow \mathrm{X}_n$$

be the inclusion map of the stratum X_n^k in X_n. We will construct the projective cover and injective envelope of IC_0 and IC_n.

Lemma 3.8. *Let $0 \leq k \leq n-1$, $U_k := \mathrm{X}_n^k \cup \mathrm{X}_n^{k+1}$ and $f_k : U_k \to \overline{\mathrm{X}_n^k}$ be the open embedding. Then,*

$$^p\mathrm{H}^0(f_{k*}(\mathrm{IC}_k\,|_{U_k})) = \mathrm{IC}_k$$
$$^p\mathrm{H}^1(f_{k*}(\mathrm{IC}_k\,|_{U_k})) = 0$$

Proof. The map $i_{k+2,k} : \overline{\mathrm{X}_n^{k+2}} \hookrightarrow \overline{\mathrm{X}_n^k}$ is the complementary closed embedding to f_k. We have a distinguished triangle

$$i_{k+2,k*}i^!_{k+2,k}\,\mathrm{IC}_k \to \mathrm{IC}_k \to f_{k*}(\mathrm{IC}_k\,|_{U_k}) \to .$$

Since IC_k is perverse, only the zeroth perverse cohomology is nonzero and

$$^p\mathrm{H}^0(\mathrm{IC}_k) = \mathrm{IC}_k\,.$$

Using 5 and the fact that $i_{k+2,k*}$ is t-exact, we get

$$^p\mathrm{H}^t(i_{k+2,k*}i^!_{k+2,k}\,\mathrm{IC}_k) = 0$$

for $t \leq 3$. Taking the long exact sequence of perverse cohomology of the above distinguished triangle, we get the desired isomorphisms. □

Lemma 3.9. *Let $0 \leq k \leq n-1$, $h_k : \mathrm{X}_n^{k+1} \hookrightarrow U_k$ be the closed embedding and $j_k : \mathrm{X}_n^k \hookrightarrow \overline{\mathrm{X}_n^k}$ be the open embedding.*

$$^p\mathrm{H}^0(j_{k+1*}h^!_k(\mathrm{IC}_k\,|_{U_k})) = 0$$
$$^p\mathrm{H}^1(j_{k+1*}h^!_k(\mathrm{IC}_k\,|_{U_k})) = {}^p\mathrm{H}^0(j_{k+1*}\underline{\Bbbk}_{\mathrm{X}_n^{k+1}}[\dim \mathrm{X}_n^{k+1}])$$

Proof. From 5, we have $h^!_k(\mathrm{IC}_k\,|_{U_k}) \in {}^p\mathrm{D}^{\geq 1}$ and j_{k+1*} is left t-exact, which implies $j_{k+1*}h^!_k(\mathrm{IC}_k\,|_{U_k}) \in {}^p\mathrm{D}^{\geq 1}$. Thus, ${}^p\mathrm{H}^0(j_{k+1*}h^!_k(\mathrm{IC}_k\,|_{U_k})) = 0$.

Using the proper base change theorem, we can write

$$\mathrm{IC}_k\,|_{U_k} \cong \pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim \mathrm{X}_n^k]|_{U_k} \cong \pi'_{k*}\underline{\Bbbk}_{\pi_k^{-1}(U_k)}[\dim \mathrm{X}_n^k]$$

where $\pi'_k : \pi_k^{-1}(U_k) \xrightarrow{\pi_k} U_k$.

The term $h^!_k\pi_{k*}\underline{\Bbbk}_{\widetilde{\mathrm{X}_n^k}}[\dim \mathrm{X}_n^k]|_{U_k} \cong h^!_k\pi'_{k*}\underline{\Bbbk}_{\pi_k^{-1}(U_k)}[\dim \mathrm{X}_n^k]$ can be further simplified using the following Cartesian square

$$\begin{array}{ccc} \pi_k^{-1}(\mathrm{X}_n^{k+1}) & \xrightarrow{h'_k} & \pi_k^{-1}(U_k) \\ \downarrow{\scriptstyle \pi''_k} & & \downarrow{\scriptstyle \pi'_k} \\ \mathrm{X}_n^{k+1} & \xrightarrow{h_k} & U_k \end{array}$$

to get

$$h^!_k\pi'_{k*}\underline{\Bbbk}_{\pi_k^{-1}(U_k)}[\dim \mathrm{X}_n^k] \cong \pi''_{k*}h'^!_k\underline{\Bbbk}_{\pi_k^{-1}(U_k)}[\dim \mathrm{X}_n^k].$$

Let ω_X be the dualizing complex for the space X. Since $\pi_k^{-1}(U_k)$ is smooth and $\dim(\pi_k^{-1}(U_k)) = \dim \mathrm{X}_n^k$, we can write

$$\begin{aligned} h_k'^{!}\underline{\Bbbk}_{\pi_k^{-1}(U_k)}[\dim \mathrm{X}_n^k] &\cong h_k'^{!}\omega_{\pi_k^{-1}(U_k)}[-\dim \mathrm{X}_n^k] \\ &\cong \omega_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[-\dim \mathrm{X}_n^k] \\ &\cong \underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[2\dim(\pi_k^{-1}(\mathrm{X}_n^{k+1})) - \dim \mathrm{X}_n^k] \\ &\cong \underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[2(\dim \mathrm{X}_n^{k+1} + k) - \dim \mathrm{X}_n^k] \\ &\cong \underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[\dim \mathrm{X}_n^{k+1} + 2k + \dim \mathrm{X}_n^{k+1} - \dim \mathrm{X}_n^k] \\ &\cong \underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[\dim \mathrm{X}_n^{k+1} - 1]. \end{aligned}$$

Since j_{k+1*} is left t-exact, we have the following

$$\begin{aligned} {}^p\mathrm{H}^1(j_{k+1*}h_k^!(\mathrm{IC}_k|_{U_k})) &\cong {}^p\mathrm{H}^1(j_{k+1*}\pi''_{k*}h_k'^{!}\underline{\Bbbk}_{\pi_k^{-1}(U_k)}[\dim \mathrm{X}_n^k]) \\ &\cong {}^p\mathrm{H}^1(j_{k+1*}\pi''_{k*}\underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[\dim \mathrm{X}_n^{k+1} - 1]) \\ &\cong {}^p\mathrm{H}^0(j_{k+1*}\pi''_{k*}\underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[\dim \mathrm{X}_n^{k+1}]) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*}\pi''_{k*}\underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[\dim \mathrm{X}_n^{k+1}]) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*}{}^p\mathrm{H}^0(\pi''_{k*}\underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[\dim \mathrm{X}_n^{k+1}])). \end{aligned}$$

The fibers of π''_k are isomorphic to $\mathbb{P}^k$. Thus, the cohomology of the fiber gives

$$\begin{aligned} {}^p\mathrm{H}^1(j_{k+1*}h_k^!(\mathrm{IC}_k|_{U_k})) &\cong {}^p\tau^{\leq 0}(j_{k+1*}{}^p\mathrm{H}^0(\pi''_{k*}\underline{\Bbbk}_{\pi_k^{-1}(\mathrm{X}_n^{k+1})}[\dim \mathrm{X}_n^{k+1}])) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*}\,\underline{\Bbbk}_{\mathrm{X}_n^{k+1}}[\dim \mathrm{X}_n^{k+1}]) \\ &\cong {}^p\tau^{\leq 0}(j_{k+1*}{}^p\mathrm{H}^0(\underline{\Bbbk}_{\mathrm{X}_n^{k+1}}[\dim \mathrm{X}_n^{k+1}])) \\ &\cong {}^p\tau^{\leq 0}j_{k+1*}(\underline{\Bbbk}_{\mathrm{X}_n^{k+1}}[\dim \mathrm{X}_n^{k+1}]) \\ &\cong {}^p\mathrm{H}^0(j_{k+1*}\underline{\Bbbk}_{\mathrm{X}_n^{k+1}}[\dim \mathrm{X}_n^{k+1}]) \end{aligned}$$

□

The following proposition will describe the composition series of some important perverse objects.

Proposition 3.10. *Let $0 \leq k \leq n-1$ and $j_k : \mathrm{X}_n^k \hookrightarrow \overline{\mathrm{X}_n^k}$ be the inclusion map. Then, we get the following exact sequences*

$$0 \to \mathrm{IC}_k \to {}^p\mathrm{H}^0(j_{k*}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k]) \to {}^p\mathrm{H}^0(j_{k+1*}\underline{\Bbbk}_{\mathrm{X}_n^{k+1}}[\dim \mathrm{X}_n^{k+1}]) \to 0$$

and

$$0 \to {}^p\mathrm{H}^0(j_{k+1!}\underline{\Bbbk}_{\mathrm{X}_n^{k+1}}[\dim \mathrm{X}_n^{k+1}]) \to {}^p\mathrm{H}^0(j_{k!}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k]) \to \mathrm{IC}_k \to 0$$

Proof: Let $g_k : \mathrm{X}_n^k \hookrightarrow U_k = \mathrm{X}_n^k \cup \mathrm{X}_n^{k+1}$ be an open embedding and $h_k : \mathrm{X}_n^{k+1} \hookrightarrow U_k$ be the complementary closed embedding. We get a distinguished triangle

$$h_{k*}h_k^!(\mathrm{IC}_k|_{U_k}) \to \mathrm{IC}_k|_{U_k} \to g_{k*}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k] \to .$$

Let $f_k : U_k \hookrightarrow \overline{\mathrm{X}_n^k}$ be the inclusion map. Applying the pushforward f_{k*} to the above distinguished triangle we get

$$j_{k+1*}h_k^!(\mathrm{IC}_k|_{U_k}) \to f_{k*}\,\mathrm{IC}_k|_{U_k} \to j_{k*}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k] \to .$$

Taking perverse cohomology, we get the following long exact sequence

$$\begin{aligned}{}^p\mathrm{H}^0(j_{k+1*}h^!_k(\mathrm{IC}_k|_{U_k})) \to {}^p\mathrm{H}^0(f_{k*}\,\mathrm{IC}_k|_{U_k}) \to {}^p\mathrm{H}^0(j_{k*}\underline{\Bbbk}_{\mathrm{X}^k_n}[\dim \mathrm{X}^k_n])\\ \to {}^p\mathrm{H}^1(j_{k+1*}h^!_k(\mathrm{IC}_k|_{U_k})) \to {}^p\mathrm{H}^1(f_{k*}\,\mathrm{IC}_k|_{U_k}) \to\end{aligned}$$

Using the Lemma 3.8 and Lemma 3.9, we get the desired result. Applying Verdier duality, we get the other short exact sequence.

Corollary 3.11. *For $0 \le k \le n$, the perverse sheaf ${}^p\mathrm{H}^0(j_{k*}\underline{\Bbbk}_{\mathrm{X}^k_n}[\dim \mathrm{X}^k_n])$ (respectively ${}^p\mathrm{H}^0(j_{k!}\underline{\Bbbk}_{\mathrm{X}^k_n}[\dim \mathrm{X}^k_n])$) is uniserial with composition factors $\mathrm{IC}_k, \cdots, \mathrm{IC}_n$ (respectively $\mathrm{IC}_n, \cdots, \mathrm{IC}_k$).*

Lemma 3.12. *The perverse sheaf ${}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n])$ (respectively ${}^p\mathrm{H}^0(j_{0*}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n])$) is the projective cover (respectively the injective envelope) of IC_0.*

Proof. Since j_0 is an open embedding, the functor j_0^* is t-exact and we have

$$\begin{aligned}&\mathrm{Hom}({}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]), {}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]))\\ &= \mathrm{Hom}({}^p\tau^{\ge 0}(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]), {}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]))\\ &= \mathrm{Hom}(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], {}^p\mathrm{H}^0\, j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n])\\ &= \mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], j_0^*({}^p\mathrm{H}^0\, j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]))\\ &= \mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], {}^p\mathrm{H}^0\, j_0^* j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n])\\ &= \mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], \underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]) \cong \Bbbk.\end{aligned}$$

Thus, the object ${}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n])$ is indecomposable.

Moreover,

$$\begin{aligned}\mathrm{Hom}((j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]), \mathrm{IC}_k[1]) &= \mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], j_0^!\,\mathrm{IC}_k[1])\\ &= \mathrm{Hom}(\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], j_0^*\,\mathrm{IC}_k[1])\end{aligned}$$

and

$$j_0^*\,\mathrm{IC}_k[1] = \begin{cases}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n + 1], & \text{if } k = 0\\ 0, & \text{otherwise}\end{cases}.$$

From the Lemma 3.5, we get $\mathrm{Hom}(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], \mathrm{IC}_k[1]) = 0$ for all $0 \le k \le n$.

Consider the distinguished triangle

$$j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n] \to {}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]) \to {}^p\tau^{<0}j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n][1] \to .$$

Applying the functor $\mathrm{Hom}(-, \mathrm{IC}_k[1])$ to the distinguished triangle, we get the following long exact sequence

$$\begin{aligned}\mathrm{Hom}({}^p\tau^{<0}j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], \mathrm{IC}_k) \to\ &\mathrm{Ext}^1({}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]), \mathrm{IC}_k)\\ &\to \mathrm{Hom}(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], \mathrm{IC}_k[1]) \to .\end{aligned}$$

Since ${}^p\tau^{<0}j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n] \in {}^p\mathrm{D}^{<0}$ and $\mathrm{IC}_k \in {}^p\mathrm{D}^{\ge 0}$, we get

$$\mathrm{Hom}({}^p\tau^{<0}j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], \mathrm{IC}_k) = 0.$$

Hence, $\mathrm{Hom}(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n], \mathrm{IC}_k[1]) = 0$ implies

$$\mathrm{Ext}^1({}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}^0_n}[\dim \mathrm{X}^0_n]), \mathrm{IC}_k) = 0.$$

Using the Proposition 3.10, we see ${}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0])$ has a unique simple quotient i.e., IC_0. Hence, it is the projective cover of IC_0. Using Verdier duality, $\mathbb{D}({}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0])) \cong {}^p\mathrm{H}^0(j_{0*}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0])$, we get the injective envelope of IC_0. □

We can also construct the injective envelope and the projective cover of IC_n.

Corollary 3.13. *Let $\mathbb{T}$ be the Fourier–Sato transform functor. Then the perverse sheaf $\mathbb{T}({}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0]))$ (respectively $\mathbb{T}({}^p\mathrm{H}^0(j_{0*}\Bbbk[\dim \mathrm{X}_n^0])))$ is the projective cover (respectively the injective envelope) of IC_n.*

Proof. The functor $\mathbb{T}$ is an equivalence of categories that preserves projectives, injectives and simples. It also takes the constant sheaf IC_0 to the skyscraper sheaf IC_n. Since ${}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0])$ is the projective cover and ${}^p\mathrm{H}^0(j_{0*}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0])$ is the injective envelope of IC_0, we get our result. □

Remark 3.14. Using the Corollary 3.11 and the Corollary 3.13, the multiplicity of IC_k in the projective cover of IC_0 and the projective cover of IC_n is one.

Using the Lemma 3.12 and the Corollary 3.11, the composition series of the projective cover of IC_0 gives us the following path in the associated quiver:

$$0 \xrightarrow{u_0} 1 \xrightarrow{u_1} 2 \xrightarrow{u_2} \cdots \xrightarrow{u_{n-1}} n.$$

Note that there is only one arrow between consecutive vertices because the multiplicity of each IC_k in the projective cover of IC_0 i.e., ${}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0])$ is one. By Verdier duality, there must also be an arrow in the opposite direction, i.e.,

$$0 \underset{v_0}{\overset{u_0}{\rightleftarrows}} 1 \underset{v_1}{\overset{u_1}{\rightleftarrows}} 2 \underset{v_2}{\overset{u_2}{\rightleftarrows}} \cdots \underset{v_{n-1}}{\overset{u_{n-1}}{\rightleftarrows}} n$$

From the Lemma 3.6 and the Lemma 3.7, there are no loops and no arrows between non-consecutive vertices. Thus, at the level of underlying quivers, we have found all the arrows.

4. Quiver with relations

Let Q be a quiver and $\Bbbk Q$ be the path algebra with basis given by directed paths in Q and multiplication induced by concatenation of paths.

Definition 4.1. A *relation* in Q is a $\Bbbk$-linear combination of paths of length at least two having the same source and target. A *quiver with relations* $\widehat{Q} := (Q, I)$ is a quiver Q with a finite set of relations I.

The quiver algebra of $\widehat{Q}$ is the quotient $\Bbbk Q/\langle I\rangle$ of the path algebra by the ideal generated by the relations. We will assume that the ideal of relations $\langle I\rangle$ contains all paths whose length is large enough, so that the corresponding quiver algebra is finite-dimensional.

Example 4.2. The quiver $\widehat{AA_n}$ is the quiver with $n+1$ vertices, arrows and relations as follows,

$$0 \underset{v_0}{\overset{u_0}{\rightleftarrows}} 1 \underset{v_1}{\overset{u_1}{\rightleftarrows}} 2 \underset{v_2}{\overset{u_2}{\rightleftarrows}} \cdots \underset{v_{n-1}}{\overset{u_{n-1}}{\rightleftarrows}} n$$

with the relations

$$u_i v_i = 0 = v_i u_i.$$

Lőrincz and Walther [LW19] showed the category of G_n-equivariant D-modules on X_n is equivalent to the category of representations of $\widehat{AA_n}$.

We have the quiver as described in the above section but we should also describe relations between the arrows u_i's and v_i's. On the level of the quiver, it matches with $\widehat{AA_n}$. We will show that it also has the same relations.

5. The main theorem

Let $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n$ be the full subcategory of $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$ consisting of G_n-equivariant perverse sheaves on X_n that do not have IC_n as a composition factor. We will try to understand this subcategory and how much information it contains to understand $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$.

Let $\mathrm{X}_{1,1}$ be the principal open subset of matrices in X_n whose $(1,1)$ entry is nonzero and

$$\mathrm{H} := \left\{ \begin{pmatrix} x & 0 \\ v & A \end{pmatrix} \times \begin{pmatrix} y & w \\ 0 & B \end{pmatrix} \;\middle|\; x, y \in \mathbb{C}^\times, v, w^T \in \mathbb{C}^{n-1} \text{ and } (A, B) \in G_{n-1} \right\}$$

be a subgroup of G_n. Observe that $\mathrm{X}_n^n = \{0\}$ is the only orbit contained entirely in the complement of $\mathrm{X}_{1,1}$. Moreover, the subset $\mathrm{X}_{1,1}$ is H-stable which gives the following:

Lemma 5.1. *Let $j : \mathrm{X}_{1,1} \hookrightarrow \mathrm{X}_n$ be the inclusion map. Then*

$$j^* : \mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n \hookrightarrow \mathrm{Perv}_{\mathrm{H}}(\mathrm{X}_{1,1}, \Bbbk) \tag{6}$$

is fully faithful.

Proof. Exactness holds because j is an open embedding. Let $F, G \in \mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n$. Let i be the complementary closed embedding to j and consider the distinguished triangle,

$$i_! i^! G \to G \to j_* j^* G \to .$$

Applying the functor $\mathrm{Hom}(F, -)$, we get

$$\mathrm{Hom}(F, i_! i^! G) \to \mathrm{Hom}(F, G) \to \mathrm{Hom}(F, j_* j^* G) \to \mathrm{Hom}(F, i_! i^! G[1]) \to .$$

Observe that ${i_*}^p \mathrm{H}^0(i^! G)$ (respectively ${i_*}^p \mathrm{H}^0(i^* F)$) is a subobject (respectively quotient) of G (respectively F), hence lies in $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$. They are supported on the complement of $\mathrm{X}_{1,1}$. The only orbit contained entirely in the complement of $\mathrm{X}_{1,1}$ is X_n^n, which implies that the only simple subobject they may contain is IC_n. Since $F, G \in \mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n$, we get

$$\begin{aligned} {i_*}^p \mathrm{H}^0(i^! G) &= 0 \\ {i_*}^p \mathrm{H}^0(i^* F) &= 0. \end{aligned}$$

This shows $i^! G \in {}^p\mathrm{D}^{\geq 1}$ and $i^* F \in {}^p\mathrm{D}^{\leq -1}$.

By adjunction, and since i is a closed embedding,

$$\mathrm{Hom}(F, i_! i^! G[k]) = \mathrm{Hom}(i^* F, i^! G[k]) = 0$$

for $k = 0, 1$. Hence,

$$\mathrm{Hom}(F, G) \cong \mathrm{Hom}(F, j_* j^* G).$$

Using adjunction for j_* i.e.,

$$\mathrm{Hom}(F, G) \cong \mathrm{Hom}(F, j_* j^* G) \cong \mathrm{Hom}(j^* F, j^* G),$$

we prove the result. □

Lemma 5.2. *There is a fully faithful functor*

$$F : \mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n \hookrightarrow \mathrm{Perv}_{\mathrm{G}_{n-1}}(\mathrm{X}_{n-1}, \Bbbk).$$

Proof. We embed X_{n-1} in $\mathrm{X}_{1,1}$ by sending $M \in \mathrm{X}_{n-1}$ to $\begin{pmatrix} 1 & 0 \\ 0 & M \end{pmatrix} \in \mathrm{X}_{1,1}$.

Set $\mathrm{G}'_n := \mathrm{G}_{n-1} \times \mathbb{C}^\times$ and embed it in H by sending $(A, B, c) \in \mathrm{G}'_n$ to $\begin{pmatrix} c & 0 \\ 0 & A \end{pmatrix} \times \begin{pmatrix} c & 0 \\ 0 & B \end{pmatrix} \in \mathrm{H}$. Since $\mathbb{C}^\times \subset \mathrm{H}$ acts trivially on the copy of $\mathrm{X}_{n-1} \subset \mathrm{X}_{1,1}$, we get

$$\mathrm{Perv}_{\mathrm{G}_{n-1}}(X_{n-1}, \Bbbk) \cong \mathrm{Perv}_{\mathrm{G}'_n}(X_{n-1}, \Bbbk). \tag{7}$$

One can check that the natural map

$$\begin{aligned} \phi : \mathrm{H} \times^{\mathrm{G}'_n} \mathrm{X}_{n-1} &\to \mathrm{X}_{1,1} \\ (h, X) &\mapsto h \cdot X \end{aligned}$$

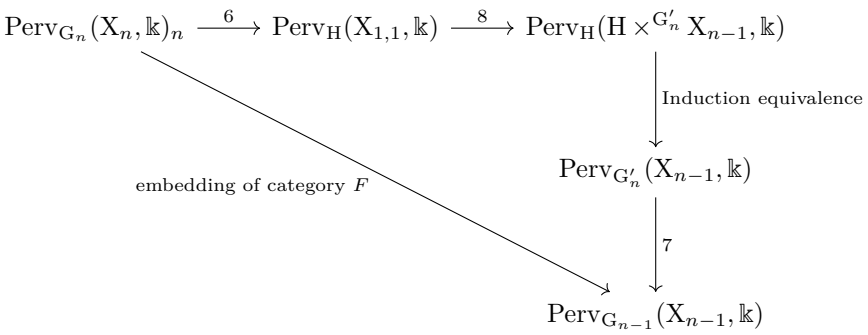
is well defined. We can also construct the inverse map

$$\begin{aligned} &\psi : \mathrm{X}_{1,1} \to \mathrm{H} \times^{\mathrm{G}'_n} \mathrm{X}_{n-1} \\ &\begin{pmatrix} x & y \\ z & A \end{pmatrix} \mapsto \left(\left(\begin{pmatrix} x & 0 \\ z & I_{n-1} \end{pmatrix}, \begin{pmatrix} 1 & -x^{-1}y \\ 0 & I_{n-1} \end{pmatrix} \right), A - x^{-1}zy \right) \end{aligned}$$

which gives an isomorphism as H-varieties, i.e.,

$$\mathrm{H} \times^{\mathrm{G}'_n} \mathrm{X}_{n-1} \cong \mathrm{X}_{1,1}. \tag{8}$$

Using the inductive equivalence [Ach21, Theorem 6.5.10], the fact that j^* is fully faithful, and the above isomorphism, we obtain an embedding of the category F as shown below



□

Theorem 5.3. *Let* $\mathrm{X}_n = \mathrm{M}_n(\mathbb{C})$ *and* $\mathrm{G}_n = \mathrm{GL}_n(\mathbb{C}) \times \mathrm{GL}_n(\mathbb{C})$. *Then there is an equivalence of categories*

$$\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk) \cong \mathit{rep}(\widehat{AA_n})$$

for all $n \geq 1$.

Proof. We will prove this statement by induction on n. It is easy to check for $n = 1$. Let $n \geq 2$ and assume $\mathrm{Perv}_{\mathrm{G}_{n-1}}(\mathrm{X}_{n-1}, \Bbbk) \cong \mathrm{rep}(\widehat{AA_{n-1}})$. The quiver associated to

$\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n$ is obtained by removing the nth vertex from the quiver associated to $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$ but keeping the same relations i.e.,

$$0 \underset{v_0}{\overset{u_0}{\rightleftarrows}} 1 \underset{v_1}{\overset{u_1}{\rightleftarrows}} 2 \underset{v_2}{\overset{u_2}{\rightleftarrows}} \cdots \underset{v_{n-2}}{\overset{u_{n-2}}{\rightleftarrows}} n-1.$$

The functor F constructed in the Lemma 5.2 is fully faithful and the categories $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n$ and $\mathrm{Perv}_{\mathrm{G}_{n-1}}(X_{n-1}, \Bbbk)$ have the same quiver. Thus, we get

$$\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)_n \cong \mathrm{Perv}_{\mathrm{G}_{n-1}}(\mathrm{X}_{n-1}, \Bbbk).$$

We get the relations

$$u_i v_i = 0 = v_i u_i$$

for $0 \leq i \leq n-2$ and no other relations amongst them.

Since $\mathbb{T}$ is an equivalence of categories sending IC_0 to IC_n and permuting the other simple objects, we get

$$\mathrm{Hom}(\mathrm{IC}_n, \mathrm{IC}_{n-1}) \cong \mathrm{Hom}(\mathrm{IC}_0, \mathrm{IC}_k).$$

for some $k \leq n-1$. The relations $(u_k \cdots u_0)(v_0 \cdots v_k) = 0 = (v_0 \cdots v_k)(u_k \cdots u_0)$ force the relation

$$u_{n-1} v_{n-1} = 0 = v_{n-1} u_{n-1}.$$

Thus, the quiver of $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$ has all the same relations as the quiver $\widehat{AA_n}$.

It remains to rule out further relations involving u_{n-1} and v_{n-1}. From the above relations, it is clear that if one is staying on some path, one cannot make a U-turn. Thus, the only possible non-zero paths which contain u_{n-1} or v_{n-1} are of the form $u_{n-1} u_{n-2} \cdots u_k = 0$ and $v_k v_{k+1} \cdots v_{n-1} = 0$. In either case, the multiplicity of IC_k in the projective cover of IC_n decreases to 0 which contradicts the Remark 3.14. Thus,

$$u_i v_i = 0 = v_i u_i$$

for $0 \leq i \leq n-1$ are all the relations in our quiver associated to $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$. □

6. Applications

The functor $\mathbb{T}$ takes the constant sheaf IC_0 to the skyscraper sheaf IC_n, but in general it is hard to compute $\mathbb{T}(\mathrm{IC}_k)$ for $0 < k < n$.

Let S^k be the simple representation in $\mathrm{rep}(\widehat{AA_n})$ associated to IC_k in $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$. Using our main result, we can try to describe indecomposable projective and injective objects from the quiver side. The projective cover of S^n is uniserial with composition factors $S^0, S^1, \ldots, S^n$. This implies that the projective cover of IC_n i.e., $\mathbb{T}({}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0]))$ is uniserial with composition factors IC_0, $\mathrm{IC}_1, \ldots, \mathrm{IC}_n$.

From 3.11, the perverse sheaf $\mathbb{T}({}^p\mathrm{H}^0(j_{0!}\underline{\Bbbk}_{\mathrm{X}_n^0}[\dim \mathrm{X}_n^0]))$ is uniserial with composition factors $\mathbb{T}(\mathrm{IC}_n), \ldots, \mathbb{T}(\mathrm{IC}_0)$. The projective cover being uniserial and matching the composition factors, we get

$$\mathbb{T}(\mathrm{IC}_k) \cong \mathrm{IC}_{n-k}\,.$$

Remark 6.1. For $0 \leq k \leq n$, using the Corollary 3.11 the perverse sheaf $\mathbb{T}({}^p\mathrm{H}^0(j_{k*}\Bbbk[\dim \mathrm{X}_n^k]))$ is uniserial with composition factors $\mathrm{IC}_{n-k}, \cdots \mathrm{IC}_0$ and $\mathbb{T}({}^p\mathrm{H}^0(j_{k!}\Bbbk[\dim \mathrm{X}_n^k]))$ is uniserial with composition factors $\mathrm{IC}_0, \cdots \mathrm{IC}_{n-k}$.

6.1. **Projective cover.** We can construct the projective cover of IC_k by understanding the properties of the projective cover from the quiver side.

Remark 6.2. In $\widehat{AA_n}$, the multiplicity of each simple S^l in the projective cover of S^k is one. The projective cover of S^k is the following representation

$$\Bbbk \xleftarrow{1} \Bbbk \xleftarrow{1} \cdots \xleftarrow{1} \Bbbk \xrightarrow{1} \cdots \xrightarrow{1} \Bbbk \xrightarrow{1} \Bbbk$$

where the source is at vertex k.

For $0 \leq k \leq n$, let

$$j_0^k : \bigsqcup_{i=0}^{k} \mathrm{X}_n^i \hookrightarrow \mathrm{X}_n$$

be the open embedding of the union of the first $k+1$ strata and

$$j_k : \mathrm{X}_n^k \hookrightarrow \mathrm{X}_n$$

be the inclusion map of the stratum X_n^k in X_n.

Lemma 6.3. *For $t < k$,*

$$\begin{aligned} {}^p\mathrm{H}^0(j_{0!}^k j_0^{k*}\,\mathrm{IC}_t) &\cong \mathrm{IC}_t \\ {}^p\mathrm{H}^0(j_{0*}^k j_0^{k*}\,\mathrm{IC}_t) &\cong \mathrm{IC}_t \\ j_{0!*}^k(j_0^{k*}\,\mathrm{IC}_t) &\cong \mathrm{IC}_t\,. \end{aligned}$$

Proof. Let $i_0^k : \bigsqcup_{i=k+1}^{n} \mathrm{X}_n^i \hookrightarrow \mathrm{X}_n$ be the closed embedding. Using the following distinguished triangle,

$$j_{0!}^k j_0^{k*}\,\mathrm{IC}_t \longrightarrow \mathrm{IC}_t \longrightarrow i_{0*}^k i_0^{k*}\,\mathrm{IC}_t,$$

we get the following long exact sequence

$${}^p\mathrm{H}^{-1}(i_{0*}^k i_0^{k*}\,\mathrm{IC}_t) \longrightarrow {}^p\mathrm{H}^0(j_{0!}^k j_0^{k*}\,\mathrm{IC}_t) \to \mathrm{IC}_t \to {}^p\mathrm{H}^0(i_{0*}^k i_0^{k*}\,\mathrm{IC}_t) \to .$$

Using 4 and t-exactness of i_{0*}^k,

$${}^p\mathrm{H}^l(i_{0*}^k i_0^{k*}\,\mathrm{IC}_t) = 0$$

for $l > -4$ which proves the first isomorphism. Applying Verdier duality to the first isomorphism, we get the second isomorphism. Using 4, 5, and [Ach21, Lemma 3.3.34] we get the third isomorphism.

□

Lemma 6.4. *Let K be a perverse sheaf such that K does not have IC_t for $t \geq k$ as a composition factor. Then,*

$${}^p\mathrm{H}^0(j_{0!}^k j_0^{k*}K) \cong j_{0!*}^k(j_0^{k*}K).$$

Proof. We will prove this using induction on the number of composition factors in K. Suppose K is simple, i.e., $K \cong \mathrm{IC}_t$. Since $t < k$, we use the Lemma 6.3.

Suppose K has n composition factors. Then there exists an exact sequence

$$0 \to \mathrm{IC}_t \to K \to K' \to 0$$

such that K' has $n-1$ composition factors. Since K does not have IC_l for $l \geq k$ as a composition factor, neither does K'. Thus, ${}^p\mathrm{H}^0(j_{0!}^k j_0^{k*}K') \cong j_{0!*}^k(j_0^{k*}K')$. Using [Ach21, Lemma 3.3.34], we prove our lemma. □

For $0 \leq k \leq n$, consider the perverse sheaves

$$\mathcal{P}_k := {}^p\mathrm{H}^0(j^k_{0!}j_0^{k*}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])))).$$

The next lemma gives the multiplicity of IC_t's in $\mathcal{P}_k$'s.

Lemma 6.5. *For $0 \leq k \leq n$ and $0 \leq t \leq n$, the multiplicity of IC_t in $\mathcal{P}_k$ is one.*

Proof. Let K be the kernel of the map $\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])) \to \mathbb{T}(\mathrm{IC}_{n-k}) \cong \mathrm{IC}_k$. Applying the right-exact functor ${}^p\mathrm{H}^0(j^k_{0!}j_0^{k*}(-))$ to the exact sequence

$$0 \longrightarrow K \longrightarrow \mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])) \longrightarrow \mathrm{IC}_k \longrightarrow 0,$$

we get a long exact sequence

$$\longrightarrow {}^p\mathrm{H}^0(j^k_{0!}j_0^{k*}K) \longrightarrow \mathcal{P}_k \longrightarrow {}^p\mathrm{H}^0(j^k_{0!}j_0^{k*}\,\mathrm{IC}_k) \longrightarrow 0.$$

From the Corollary 3.11 and the Remark 6.1, the kernel K does not contain IC_t for $t \geq k$, and using the Lemma 6.4 we also get exactness on the left, i.e.,

$$0 \longrightarrow j^k_{0!*}j_0^{k*}K \longrightarrow \mathcal{P}_k \longrightarrow {}^p\mathrm{H}^0(j^k_{0!}j_0^{k*}\,\mathrm{IC}_k) \longrightarrow 0$$

is an exact sequence.

Using the Remark 6.1, the kernel K has composition factors IC_t for $t < k$ with multiplicity one. Using the Lemma 6.3, we get that the multiplicity of IC_t in $j^k_{0!*}j_0^{k*}K$ for $t < k$ is one.

Using the Corollary 3.11, the object

$${}^p\mathrm{H}^0(j^k_{0!}j_0^{k*}\,\mathrm{IC}_k) \cong {}^p\mathrm{H}^0(j_{k!}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k])$$

has composition factors IC_t for $t \geq k$ with multiplicity one. Thus the multiplicity of IC_t for $0 \leq t \leq n$ in $\mathcal{P}_k$ is one. □

Lemma 6.6. *For $0 \leq k \leq n$, the perverse sheaf $\mathcal{P}_k$ has a unique simple quotient IC_k.*

Proof. Observe that for $0 \leq t \leq n$, we have

$$\begin{aligned}
\mathrm{Hom}(\mathcal{P}_k, \mathrm{IC}_t) &\cong \mathrm{Hom}({}^p\mathrm{H}^0(j^k_{0!}j_0^{k*}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])))), \mathrm{IC}_t) \\
&\cong \mathrm{Hom}({}^p\tau^{\geq 0}(j^k_{0!}j_0^{k*}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])))), \mathrm{IC}_t) \\
&\cong \mathrm{Hom}(j^k_{0!}j_0^{k*}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}]))), \mathrm{IC}_t) \\
&\cong \mathrm{Hom}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])), j^k_{0*}j_0^{k*}\,\mathrm{IC}_t) \\
&\cong \mathrm{Hom}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])), {}^p\mathrm{H}^0(j^k_{0*}j_0^{k*}\,\mathrm{IC}_t)).
\end{aligned}$$

For $t > k$, we have $j^k_{0!}j_0^{k*}\,\mathrm{IC}_t = 0$ which implies

$$\mathrm{Hom}(\mathcal{P}_k, \mathrm{IC}_t) = 0.$$

For $t < k$, we have ${}^p\mathrm{H}^0(j^k_{0*}j_0^{k*}\,\mathrm{IC}_t) \cong \mathrm{IC}_t$. Thus,

$$\mathrm{Hom}(\mathcal{P}_k, \mathrm{IC}_t) \cong \mathrm{Hom}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\Bbbk[\dim \mathrm{X}_n^k])), \mathrm{IC}_t) = 0$$

since $\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}]))$ has a unique simple quotient IC_k.

For $t = k$, we have

$$\mathrm{Hom}(\mathcal{P}_k, \mathrm{IC}_k) \cong \mathrm{Hom}(\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}])), {}^p\mathrm{H}^0(j_{k*}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k])).$$

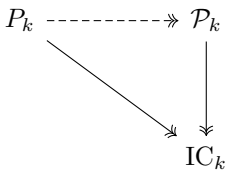
Using the Corollary 3.11 and the Remark 6.1, $\mathbb{T}({}^p\mathrm{H}^0(j_{n-k!}\underline{\Bbbk}_{\mathrm{X}_n^{n-k}}[\dim \mathrm{X}_n^{n-k}]))$ has a unique simple quotient IC_k and IC_k is the unique simple subobject of ${}^p\mathrm{H}^0(j_{k*}\underline{\Bbbk}_{\mathrm{X}_n^k}[\dim \mathrm{X}_n^k])$. Since IC_k appears in both with multiplicity one, we get $\mathrm{Hom}(\mathcal{P}_k, \mathrm{IC}_k) = \Bbbk$.

□

The following theorem proves $\mathcal{P}_k$ is the projective cover of IC_k.

Theorem 6.7. *For $0 \leq k \leq n$, the perverse sheaf $\mathcal{P}_k$ is the projective cover of IC_k in $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$.*

Proof. Let P_k be the projective cover of IC_k in $\mathrm{Perv}_{\mathrm{G}_n}(\mathrm{X}_n, \Bbbk)$. From the Lemma 6.6, the perverse sheaf $\mathcal{P}_k$ has a unique simple quotient IC_k which gives a surjective map from P_k to $\mathcal{P}_k$.



Using the Remark 6.2 and the Lemma 6.5, the multiplicity of simples matches in both perverse sheaves. The surjective map between perverse sheaves with the same composition factors is an isomorphism which proves our theorem. □

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